

On the Intelligent Control Design of an Agricultural Mobile Robot for Cotton Crop Monitoring

Adalberto I. S. Oliveira
Dept. of Electrical Engineering
PUC-Rio
Rio de Janeiro, Brazil
adalberto@ele.puc-rio.br

Thiago M. Carvalho
Dept. of Electrical Engineering
PUC-Rio
Rio de Janeiro, Brazil
tmedeiros@ele.puc-rio.br

Felipe F. Martins
Dept. of Electrical Engineering
PUC-Rio
Rio de Janeiro, Brazil
fferraz@ele.puc-rio.br

Antonio C. Leite
Dept. of Electrical Engineering
PUC-Rio
Rio de Janeiro, Brazil
antonio.cleite@ele.puc-rio.br

Karla T. Figueiredo
Dept. of Informatics and Computer Science
Rio de Janeiro State University
Rio de Janeiro, Brazil
karlafigueiredo@ime.uerj.br

Marley M. B. R. Vellasco
Dept. of Electrical Engineering
PUC-Rio
Rio de Janeiro, Brazil
marley@ele.puc-rio.br

Wouter Caarls
Dept. of Electrical Engineering
PUC-Rio
Rio de Janeiro, Brazil
wouter@ele.puc-rio.br

Abstract—In this work, we address the modelling and control design of a wheeled mobile robot capable of performing autonomous navigation tasks in agricultural fields. The methodology is based on the kinematics approach due to its well-known ability to ensure satisfactory performance when the robot motions are carried out with low velocities and slow accelerations. Two control strategies are used for stabilization and trajectory tracking purposes: the first scheme is based on a fuzzy logic algorithm and considers the regulation problem of the robot position in Cartesian space; the second scheme is based on a visual servoing algorithm and considers the tracking problem of a straight line by using a constant linear velocity in Cartesian space and the position error in image space. Simulation results are presented to illustrate the effectiveness and feasibility of the fuzzy logic approach. Experimental tests with an agricultural mobile robot are carried out to verify and validate the visual servoing approach for cotton crop monitoring.

Index Terms—agricultural robotics, visual servoing, fuzzy logic, computer vision, kinematic control.

I. INTRODUCTION

The agricultural sector has demanded innovative technologies that allow the increase of production, such as autonomous machines, genetically modified seeds, sensors for environmental monitoring, among others. As a result, lower production costs and reduction of environmental impacts have become relevant topics for research. In cotton and soybeans crops, for instance, most of the production costs are related to pest, weed and disease controls, which are based on visual monitoring with workers previously trained as well as the use of pesticides and herbicides [1]. According to agriculture consultancy companies in Brazil, the expenditure on pesticides and herbicides in the states of Mato Grosso and Bahia, the

largest Brazilian cotton producers, can reach up to 34% of the total production cost. To reduce such impressive costs, new techniques for crop maintenance and sustainability are being evaluated such as ultraviolet radiation for inactivation of microorganisms [2], laser-based weed control, biological pest control and autonomous mobile robots for visual monitoring of pests [3]. Taking into account that most of the field tasks are tedious and repetitive, the use of autonomous robotic systems becomes a solution to reduce the manpower usage, increase the productivity and optimize the agricultural processes. The latest developments of the Thorvald platform show us how robots can be used for transportation, UV treatment, phenotyping, grass cutting, soil sampling, precision weeding, soft fruit picking, and controlling mildew with light treatment [4]. In this context, mobile robots with hybrid weeding mechanisms are being manufactured and endowed with vision-based capabilities for online weed detection and classification, allowing for specialized treatment of individual species [5].

An automatic guidance system for a four-wheel-steering mobile robot is developed in [6] to perform accurate field operations based on the full compensation of sliding effects during path tracking. In [7], an adaptive steering system based on a model reference adaptive control (MRAC) is designed to control the lateral position of a farm tractor with varying yaw rate properties in order to track a straight path. A skid steering mobile robot was developed in [8] to infiltrate into tunnels of dense vegetation that occur in sugarcane plantations, performing mapping and collecting samples.

In this work, we present an intelligent control design for autonomous navigation of a wheeled mobile robot to monitor

cotton and soybeans crops, as shown in Fig. 1(a). The modelling approach is based on the robot kinematics which yields satisfactory performance when the task motions do not require too fast speeds or rapid accelerations. Two control strategies are used to tackle the stabilization and tracking problems: the first is based on a fuzzy logic algorithm ensuring a successful regulation of the robot posture in the Cartesian space; the second is based on a visual servoing approach and considers the following of a straight line by using a constant linear velocity in the task space and the position error in the image space. Numerical simulations are included to illustrate the effectiveness and feasibility of the fuzzy logic algorithm. The verification and validation of the visual servoing approach is performed through experimental tests with a differential-drive mobile robot monitoring cotton crops, as depicted in Fig. 1(b).

II. PROBLEM STATEMENT

In the agricultural industry, automation and mechanization allow for the cultivation of gigantic areas, which demands minimally structured environments, such as plain terrain, geo-referencing, and alignment of the plantation (Fig. 1(a)). This type of infrastructure contributes to the use of autonomous machinery as well as mobile robotic system, such rovers (Fig. 1(b)) or drones, that can use the row crops as guide for localization and navigation purposes [9], [10].



Fig. 1: (a) Superior view of a structured cotton plantation; (b) SoyBot, a differential-drive mobile robot able to perform autonomous navigation through row crops.

In this context, the control problem consists in guiding a mobile robot through the plantation using crop rows as a reference path, in the same way as a line follower robot (e.g., AGV systems). To perform this task, the first approach is to use the odometric localization to estimate the current robot configuration in order to achieve a desired configuration. The key idea is to generate a straight line on a plane, minimize the robot normal distance from the line and adjust the heading angle of the vehicle to keep it parallel to the line. The second approach is to use a single camera as an input sensor [11], providing the desired orientation to the mobile robot, while the forward velocity remains constant. The line following task is useful in data collection routines, such as soil sample collection, image dataset creation, environmental conditions monitoring, or in monitoring tasks such as weed detection [10], where the robot has to move through the plantation in an autonomous mode.

A. Robot Modelling

This section presents the mathematical modelling which defines the behaviour of unicycle-like mobile robots. The unicycle robot is considered as a rigid body with wheels on a plane represented by the generalized coordinates vector $q \in \mathbb{R}^3$ defined as $q = [x \ y \ \theta]^T$, where $(x, y) \in \mathbb{R}^2$ are the Cartesian position on the plane and $\theta \in \mathbb{R}$ is the orientation of the x -axis of the robot frame $\mathcal{F}_r$ with respect to the x -axis of the world frame $\mathcal{F}_w$. In this context, the unicycle robot is a vehicle with a single orientable wheel and its kinematic model can be described by [12]:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta \\ \sin \theta \\ 0 \end{bmatrix} v + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \omega, \quad (1)$$

where the inputs $v \in \mathbb{R}$ and $\omega \in \mathbb{R}$ are the driving velocity and the steering velocity, respectively. However, a single-wheeled mobile robot has a serious balance problem under static conditions. Thus, such a vehicle can be replaced by a differential-drive mobile robot, which is more stable from the mechanical point of view. In this case, the contribution of each wheel is taken into account to compute both linear and angular velocities [12]. Then, the correspondence between the set of control inputs $(v, \omega) \in \mathbb{R}^2$ and the spinning velocities of right and left wheels, denoted by $\dot{\phi}_R \in \mathbb{R}$ and $\dot{\phi}_L \in \mathbb{R}$, is given by

$$v = \frac{r(\dot{\phi}_R + \dot{\phi}_L)}{2}, \quad \omega = \frac{r(\dot{\phi}_R - \dot{\phi}_L)}{d}, \quad (2)$$

where $r > 0$ is the radius of the wheels and $d > 0$ is the distance between the wheels centres. It is worth mentioning that the pure rolling constraint for the wheels is given by $\dot{x} \sin \theta - \dot{y} \cos \theta = 0$, which means that the velocity of the wheel centre is zero in the direction orthogonal to the sagittal axis of the vehicle. Fig. 2 shows the representation of the mobile robot, its generalized coordinates, wheels and frames of interest.

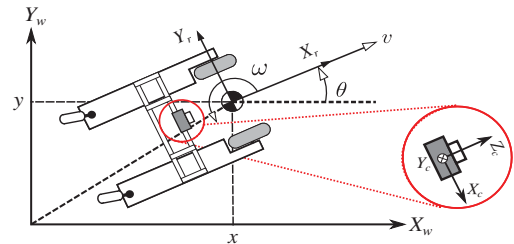


Fig. 2: Robot kinematics and its frames of interest.

B. Camera-Robot System Model

Consider a pin-hole camera moving with the body velocity $(V, \Omega) \in \mathbb{R}^6$ and observing an arbitrary point $\mathcal{P}$ located in the agricultural environment with camera frame coordinates $p_c = (x_c, y_c, z_c) \in \mathbb{R}^3$. The velocity of the point $\mathcal{P}$ with respect to the camera frame $\mathcal{F}_c$ is given by

$$\dot{p}_c = -\Omega \times p_c - V, \quad (3)$$

where $V = (V_x, V_y, V_z) \in \mathbb{R}^3$ and $\Omega = (\Omega_x, \Omega_y, \Omega_z) \in \mathbb{R}^3$ are the linear and angular velocities of the camera in the world

frame $\mathcal{F}_w$ respectively. The point $\mathcal{P}$ can be represented in the image space using the following relationship:

$$\begin{bmatrix} x_v \\ y_v \end{bmatrix} = \frac{f}{z_c} \begin{bmatrix} \alpha_x & 0 \\ 0 & \alpha_y \end{bmatrix} \begin{bmatrix} x_c \\ y_c \end{bmatrix} + \begin{bmatrix} x_{v0} \\ y_{v0} \end{bmatrix}, \quad (4)$$

with $(x_v, y_v) \in \mathbb{R}^2$ being the coordinates of the point $\mathcal{P}$ expressed in the image frame $\mathcal{F}_v$, $f > 0$ is the focal length of the camera lens, $\alpha_x, \alpha_y > 0$ are the camera scaling factors, $(x_{v0}, y_{v0}) \in \mathbb{R}^2$ are the coordinates of the principal point. Here, without loss of generality, we assume that: (A1) the mobile robot operates in the xy -plane and rotates only about z -axis of the world frame $\mathcal{F}_w$; (A2) the camera is attached to the robot structure so that its optical axis is parallel to the sagittal axis of the vehicle. In this context, substituting (3) into the time-derivative of (4) and considering only the permissible motions, we obtain $(V_z, -V_x, -\Omega_y) = (\dot{x}, \dot{y}, \dot{\theta})$ which consequently yields:

$$\begin{bmatrix} \dot{x}_v \\ \dot{y}_v \end{bmatrix} = \begin{bmatrix} \frac{\bar{x}_v}{z_c} & \frac{f\alpha_x}{z_c} & \frac{\bar{x}_v^2}{f\alpha_x} + f\alpha_x \\ \frac{\bar{y}_v}{z_c} & 0 & \frac{\bar{x}_v\bar{y}_v}{f\alpha_x} \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}, \quad (5)$$

where $\bar{x}_v = x_v - x_{v0}$ and $\bar{y}_v = y_v - y_{v0}$. Notice that, (5) relates the velocity of the point $\mathcal{P}$ in the image frame $\mathcal{F}_v$ with the velocity of the camera in the world frame $\mathcal{F}_w$.

III. CONTROL DESIGN

In this section, we present two approaches to control the mobile robot for autonomous navigation in row crops, namely a fuzzy logic control and a visual based control. In the first case, we formulate the navigation problem as a partial regulation task. For the last case, the key idea is to follow a line in the image space keeping the alignment between the robot sagittal axis with the crop rows.

A. Fuzzy Control for Regulation Task

The navigation task can be formulated as a regulation problem, wherein it is not always necessary to obtain an accurate measurement of the distance to the goal in order to design the driving action. Due to the fact that fuzzy logic approaches can handle inaccuracies and uncertainties, Fuzzy Inference Systems are commonly used in control applications, such as quadrotor control [13] and autonomous navigation [14]. Here, a fuzzy logic control approach is presented as an alternative to conventional controllers with fixed parameters. In this case, the control goal is to drive the mobile robot from an initial configuration $q_0 = [x_0 \ y_0 \ \theta_0]^T \in \mathbb{R}^3$ to a desired configuration $q_d = [x_d \ y_d \ \theta_d]^T \in \mathbb{R}^3$.

Based on the unicycle kinematic model (1), it is necessary to design two fuzzy controllers: one for the linear velocity v and other for the angular velocity ω . To achieve this aim, we have to define the fuzzy parameters for each controller in order to implement them. Thus, for the linear velocity v , the inputs are the distance error between the current position and the goal e_d and its time-derivative $\dot{e}_d$. Similarly, for the angular velocity ω , the inputs are the angular error between

the current orientation and the vector pointing to the goal e_a and its time-derivative $\dot{e}_a$. All these inputs, as well as the outputs, are considered linguistic variables in the Fuzzy Logic approach and can be defined as:

$$e_d = \sqrt{(x_d - x_0)^2 + (y_d - y_0)^2}, \quad (6)$$

$$e_a = \text{atan2}((y_d - y_0), (x_d - x_0)). \quad (7)$$

For linear velocity control, we define the fuzzy sets for three variables: the distance error e_d , the time-derivative of the distance error $\dot{e}_d$ and the linear velocity v . Since more accurate control of the robot motions is necessary when the distance error e_d is close to zero, we define seven fuzzy sets for this variable: Far (F), Medium-Far (MF), Medium (M), Medium-Close (MC), Close (C), Really Close (RC) and Really-Really-Close (RRC). For the time-derivative of the distance error $\dot{e}_d$, we define five fuzzy sets: Negative (N), Little Negative (LN), Zero (Z), Little Positive (LP) and Positive (P). Finally, for the linear velocity v , we define four fuzzy sets: Very Positive (VP), Positive (P), Little Positive (LP) and Zero (Z) since we assume only forward motions. Fig. 3 (a), (c), and (e) show the membership functions for this fuzzy logic control design.

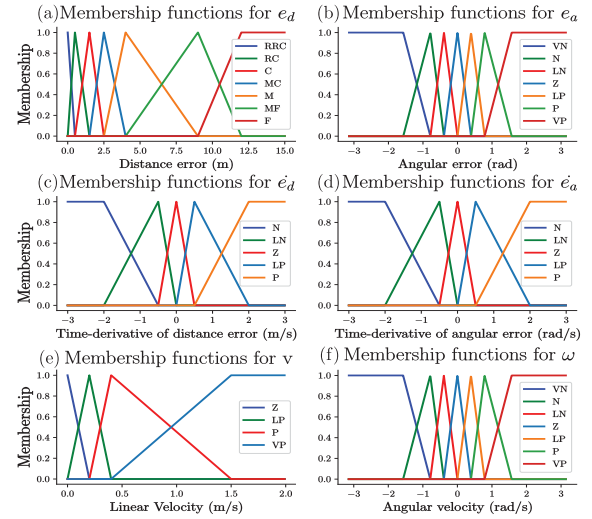


Fig. 3: Fuzzy membership functions.

In the same way as for linear velocity, for the angular velocity control we create fuzzy sets for each linguistic variable. For the angular error e_a , the fuzzy sets defined are: Very Negative (VN), Negative (N), Little Negative (LN), Zero (Z), Little Positive (LP), Positive (P) and Very Positive (VP). For the time-derivative of the angular error $\dot{e}_a$, we define the same fuzzy sets as the time-derivative of the distance error $\dot{e}_d$. Lastly, for the angular velocity ω we define seven fuzzy sets: Very Positive (VP), Positive (P), Little Positive (LP), Zero (Z), Little Negative (LN), Negative (N) and Very Negative (VN). Fig. 3 (b), (d), and (f) show the membership functions for this fuzzy control design. For linear velocity control, we create rules based on the trajectory of the mobile robot. In the same way, for angular velocity control, we create rules

based on the orientation of the mobile robot (see Tables I and II). These rules aim to correct the orientation, moving the robot to the left or right direction, depending on the sign and magnitude of the angular error e_a and its time-derivative $\dot{e}_a$. Finally, we use the well-known Centroid method for carrying out the defuzzification process, as presented in [15].

TABLE I: Rule Base for linear velocity control.

dDistError	distError						
	F	MF	M	MC	C	RC	RRC
N	VP	VP	P	P	LP	LP	Z
LN	VP	P	P	LP	LP	LP	Z
Z	P	LP	LP	LP	LP	Z	Z
LP	LP	LP	LP	LP	Z	Z	Z
P	LP	LP	LP	LP	Z	Z	Z

TABLE II: Rule Base for angular velocity control.

dAngError	angError						
	VN	N	LN	Z	LP	P	VP
N	VN	N	LN	Z	LP	P	VP
LN	VN	LN	LN	Z	LP	LP	VP
Z	N	LN	LN	Z	LP	LP	P
LP	N	N	N	Z	P	P	P
P	N	N	N	Z	P	P	P

After all rules have been created, the fuzzy logic control was implemented using the SciKit-Fuzzy library in Python language. The numerical simulations consist of the autonomous motion of the mobile robot from an initial configuration $q_0 = [1 \ 1 \ 0]^T$ to the desired position located at the origin $q_d = [0 \ 0 \ \theta_d]^T$. In this case, for the sake of simplicity, we consider only the Cartesian regulation problem and the desired orientation θ_d was not taken into account. The partial regulation task can be of practical interest when a mobile robot is exploring an uncertain environment and has to visit a sequence of checkpoints from where it perceives the features of the surrounding area using its on-board sensors [12]. Fig. 4 shows the results for this simulation and the behavior over time of the system variables.

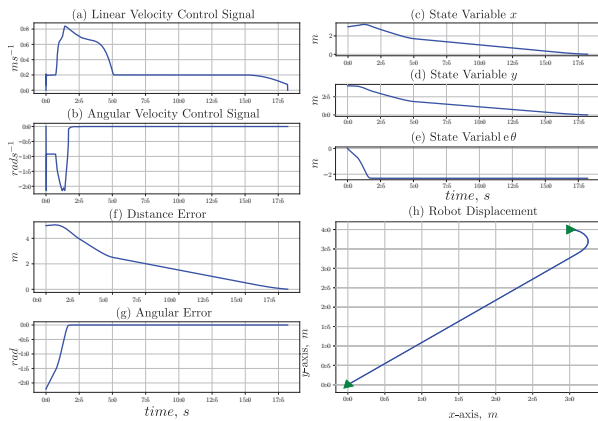


Fig. 4: Fuzzy controller results: (a) linear and (b) angular velocities; state variables (c) x , (d) y and (e) θ ; (f) distance and (g) angular errors; (h) robot displacement in xy plane.

B. Visual Servoing for Orientation Correction

Consider that the control goal is to keep the image centroid coordinates (x_v, y_v) , obtained by an appropriate image segmentation technique, aligned with the image center point (x_{vd}, y_{vd}) while the robot linear velocity v remains constant. Then, substituting (1) into (5) and considering $v = v_d$ we obtain the following vision-based control system:

$$\dot{x}_v = \frac{\bar{x}_v}{z_c} v_d \cos \theta + \frac{f \alpha_x}{z_c} v_d \sin \theta + \left(\frac{\bar{x}_v^2}{f \alpha_x} + f \alpha_x \right) \omega, \quad (8)$$

$$\dot{y}_v = \frac{\bar{y}_v}{z_c} \cos \theta + \frac{\bar{x}_v \bar{y}_v}{f \alpha_x} \omega. \quad (9)$$

Here, without loss of generality, we consider that during the robot motions through the crop row, there is no motion of the point $\mathcal{P}$ along the y -axis of the image frame $\mathcal{F}_v$ and its depth coordinate z_c remains constant. Under these assumptions, the control system (8)-(9) takes the form:

$$\dot{x}_v = \frac{f \alpha_x}{z_c} v_d \sin \theta + f \alpha_x \omega. \quad (10)$$

The visual servoing problem is to ensure that the mobile robot visually follows the image target, keeping its orientation aligned to the row crop. In this context, the control goal can be simply defined by:

$$x_v \rightarrow x_{vd}, \quad e_v = x_{vd} - x_v \rightarrow 0, \quad (11)$$

where $e_v \in \mathbb{R}$ is the image error in x -axis of the image frame $\mathcal{F}_v$ and x_{vd} is assumed to be constant or zero. Then, in order to guarantee the stabilization of the image error e_v to zero, we propose the following control law:

$$\omega = \frac{k_v}{f \alpha_x} e_v + \frac{v_d}{z_c} \sin \theta, \quad k_v > 0, \quad (12)$$

where $k_v > 0$ is the proportional gain. Notice that, all necessary signals to implement the control law (12) are available since x_v is measured by the camera and the vehicle's orientation θ can be updated by using odometry or a compass.

IV. EXPERIMENTAL RESULTS

In this section, we present the experimental results of the visual servoing control applied to the differential-drive mobile robot. The experiments were performed in a cotton farm, which is a structured environment as described in Section II, with flat terrain and aligned row crops with a clear path between them (Fig. 1). The robot used for the tests, the SoyBot, was developed by PUC-Rio in partnership with Solinftec. The kinematic model of the robot is equal to the differential-drive robot and unicycle models described in Section II-A. The robot is equipped with three WebCams Logitech C270 HD (720p and 30 fps) for image-based navigation, whose frames are aligned to the robot frame $\mathcal{F}_r$: one on top, pointing to the ground, and two above each wheel drive, pointing forward. The communication protocol for integrating the sensors was implemented on the Robot Operating System (ROS) framework, version Melodic Morenia, running on the Linux OS Ubuntu 18.04 LTS by using an Intel NUC7i5BNK Mini PC,

Intel Core i5-7260U 2.2GHz, 16GB DDR4 RAM. The ROS nodes to visually control the robot motion were developed in Python programming language using OpenCV.

For the row crop navigation task, the information that needs to be extracted from the RGB image, captured by the camera, is the position of an arbitrary point $\mathcal{P}$ (midpoint) located between two rows where the vehicle is moving. From this information, the robot is able to perform alignment maneuvers while moving through the plantation. One way to obtain such information is to use an image segmentation technique and extract the centroid of the region of interest (RoI). The proposed navigation task in open fields demands a robust segmentation algorithm in order to deal with changing light conditions. A robust solution proposed by Wei *et al* [16] uses a combination of two techniques: an RGB color space transformation to the OHTA color space and an adaptive threshold algorithm based on the image histogram analysis. The OHTA color space is obtained from a linear conversion between the RGB color space into an orthogonal color feature set (or channels) I_i and I'_i for $i=1, 2, 3$, which is given by:

$$\begin{aligned} I_1 &= (R + G + B)/2, & I'_1 &= R - G, \\ I_2 &= (R - B)/2, & I'_2 &= R - B, \\ I_3 &= (2G - R - B)/4, & I'_3 &= (2G - R - B)/2. \end{aligned} \quad (13)$$

Notice that, the original RGB color space uses three color channels (RGB) per image feature, while OHTA color space uses a grayscale channel which reduces the computational cost.

The next step consists of creating a binary image, where is applied the Otsu method, based on an image histogram analysis for dynamic class separation. The key idea is to compute the threshold value for binarization based on the quotient between the biggest value of between-class variance and the sum of this value and the inner-class smallest value (collectively variance). The solution proposed by [16] was used for fruit detection, where the objective was to extract features from ripe fruits in complex backgrounds. For this case, the channels of the OHTA color space, I_2 , I'_1 and I'_3 , was chosen due to the predominance of the red color tones. For the row crop navigation task, the predominant color tones are yellow (floor and ground) and green (leaf and plant). The channels that presented the best results was I'_1 and I'_3 . To improve the segmentation process, a Gaussian Blur filter for smoothing and noise reduction was applied to OHTA channels before the binarization process using the Otsu method. The arbitrary point $\mathcal{P}$ is given by the centroid coordinates (x_v, y_v) of the biggest area in the binary image (see Fig. 5(c)), on which is computed its displacement (e_v) with respect to the center of the image $(x_{vd}=0)$. The steps of the image feature extraction process are shown in Fig. 5.

The results correspond to one lap, with elapsed time of 433 seconds (i.e., about 7 minutes) and traveling distance of approximately 30 meters. At the beginning of the experiments, both world and robot frames, $\mathcal{F}_w$ and $\mathcal{F}_r$, are assumed to be coincident and the initial robot configuration $q(0)$ is assumed to be zero. Since the x -axis of the robot frame $\mathcal{F}_r$ is not

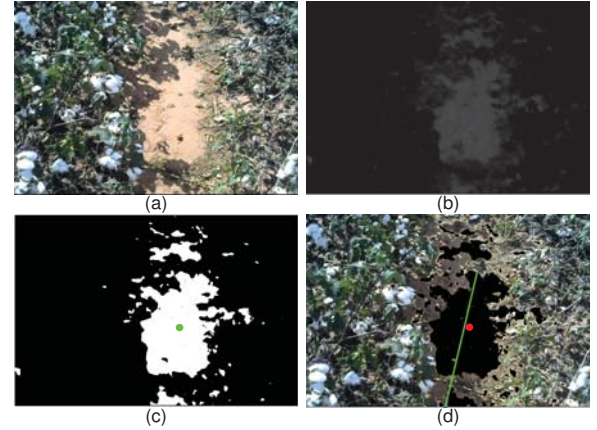


Fig. 5: Steps of crop row alignment: (a) raw image; (b) OHTA channel I'_1 ; (c) Binary image (mask) and its centroid; (d) estimated misalignment of the mobile robot.

aligned to the row crops, the mobile robot needs to perform alignment maneuvers during the navigation task. Figures 6(a)-(f) show the behavior over time of such maneuvers from the robot point of view by using only the camera mounted on top of the robot structure, with a narrow field of view (FoV), which allows only the row crop visualization. This behaviour can be seen in Fig. 7(a) wherein the robot moves along the row crop in a straight line with heading angle around 0.2 rad , as shown in Fig. 7(c). As expected, the robot was able to perform the autonomous navigation task, following the rows satisfactorily. Fig. 8 shows the control signals v, ω applied to the robot and its estimated velocities based on the readings of the wheel encoders. Figure 9 depicts the velocity commands $\dot{\phi}_R, \dot{\phi}_L$ applied to each wheel drive controller and the estimated wheel speeds based on the encoder readings.

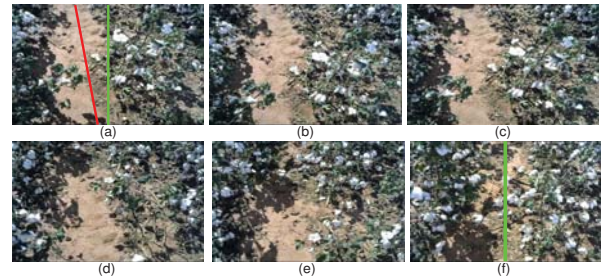


Fig. 6: Top view of the alignment maneuvers performed by the SoyBot robot.

V. CONCLUDING REMARKS

In this paper, we have proposed an intelligent control system for a differential-drive mobile robot by using an image-based visual servoing approach, which is able to autonomously drive the vehicle through the row crops. We also have presented the robot modelling, a fuzzy logic-based controller and an image segmentation technique based on a color space transformation and an adaptive threshold binarization method. Numerical

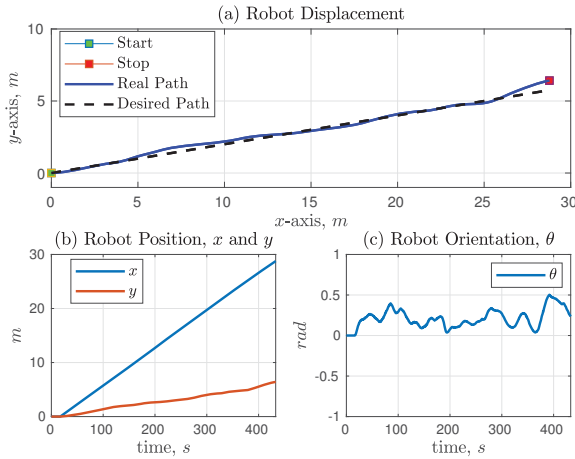


Fig. 7: (a) Robot displacement; (b) robot position in x - and y -axis; (c) robot orientation, θ .

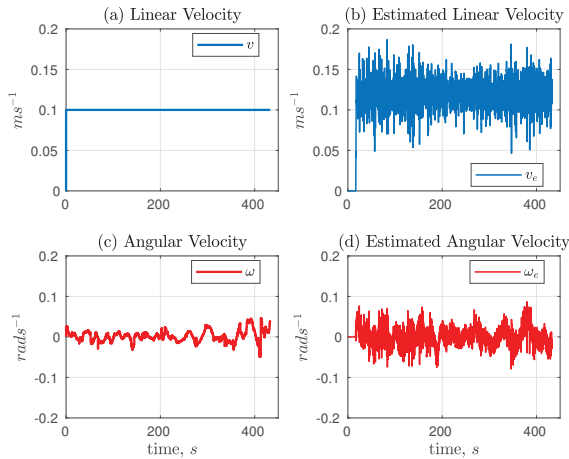


Fig. 8: (a)-(c) Linear and angular velocities commands; (b)-(d) estimated linear and angular velocities.

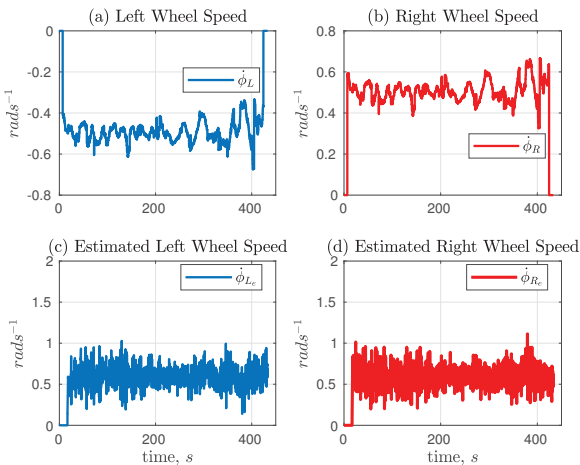


Fig. 9: (a)-(b) Left and right wheel speeds commands; (c)-(d) estimated left and right wheel speeds.

simulations show the feasibility of the fuzzy logic approach for regulation tasks. Experimental results, obtained with the SoyBot robot navigating in a real cotton field, were presented in order to validate the proposed control methodology. As future work, we will further implement the proposed fuzzy logic controller to compare it with the image-based visual servoing approach, evaluating their robustness to parametric uncertainties and external disturbances. A sensor fusion method based on the Kalman filter approach will be also developed to obtain the best midpoint $\mathcal{P}$ between the rows using the other two cameras which are mounted above the robot wheels.

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